



DIGITAL TECHNOLOGIES

ZUKUNFT. DIGITAL. NACHHALTIG.

SoSe 2026

Vergabepilot.AI:

Agentic AI for Automated Public Project Scraping

In Cooperation with



Machine Learning and COgnitive SoftwaRE (CORE) Research Group
European AI Team Projects (ATP)

Ein gemeinsames Studienprogramm der



 [digitecstudieren.de](https://www.instagram.com/digitecstudieren.de)

Project in Cooperation with a fast-moving startup!

Ciconia Systems GmbH

We develop custom software solutions and AI applications for medium-sized companies. Our focus is on the rapid implementation of digital products using modern technologies—from web apps to generative AI.



Focus
AI-
Applications



Location
Stuttgart &
Cluj



Founded in
2022



Team
20+



1.Hackathon (1)

Phase 1: Manual Scraping (5h)

- Learn manual scraping techniques
- XML Path Language (XPath)
- Browser automation (e.g., *Playwright*)
- Two teams, 7,500 URLs per team
- Implement scrapers manually (**no LLMs**)

Goal: Maximize number of scraped websites

Phase 1
Manual Scraping



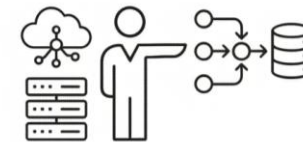
XPath, browser automation

Phase 2: LLM-Assisted Scraping

- Use different LLMs per team
- Generate scraper code with LLMs
- Test, debug, and iterate
- Keep track of iterations necessary
- How good are LLMs at generating?

Goal: Initial evaluation of LLM performance

Phase 2
Advanced & Automated Scraping



Headless browser clusters, structured data output, database loading, error handling

1.Hackathon (2)

	id	created	updated	version	url	domain	trigger_id	procedure_id
0	6a05934a-d751-4107-a1dd-dbb24ed438f7	2026-03-22 07:17:14.373 +0200	2026-03-22 07:17:25.867 +0200	1	https://www.evergabe-online.de/tenderdetails.h...	www.evergabe-online.de	b32d2a72-368a-45fd-a1f3-ef5b7aa4c101	02e5e8c9-2b40-40d0-8bd1-8b7d55f686aa
1	1342d4ba-c967-473a-8418-1f9091a813e0	2026-03-22 07:17:04.622 +0200	2026-03-22 07:17:04.983 +0200	1	https://www.dtv.de/Satellite/notice/CXP4Y89M9...	www.dtv.de	39e68c6b-1a43-487a-85af-41de8e61ef3e	d5f9a789-3e3a-4daf-b98f-469d248586dd
2	5bd3fef0-b34c-4396-94bf-b63ff81039bd	2026-03-22 07:16:31.850 +0200	2026-03-22 07:16:55.100 +0200	1	https://www.sachsen-vergabe.de/NetServer/Publi...	www.sachsen-vergabe.de	5482a1d3-1edb-42af-aff3-0a1a11970c75	a33147d0-24d4-4b23-a19f-ddad82695c16
3	e4e63f2b-6679-488e-8349-0cdbc980dae1	2026-03-22 07:16:10.828 +0200	2026-03-22 07:16:24.689 +0200	1	https://www.evergabe.de/auftraege/suche-ueber-...	www.evergabe.de	d107d276-156c-4227-add6-9347900d62a0	47f9c70e-9eba-42df-a432-19fb24b6b1c6
4	504d3c99-f8e8-4f35-8de0-3d8afce25f7a	2026-03-22 07:15:47.255 +0200	2026-03-22 07:15:59.975 +0200	1	https://vergabeportal-bw.de/Satellite/notice/C...	vergabeportal-bw.de	cfbae5e1-472a-4154-a475-dade4f3c76a3	ed5c05af-a15f-42b9-bb48-5237aca5498d

Scraper Dataset

- One scraper can be used across the same platform / websites
- Does your scraper work well on other examples of the same website?
- Older publications might not be available anymore (invalid)

OpenRouter API keys are available (50\$ starting budget per team)

2. Final Week Goals

Objective

- Explore computer-use agents (CUA)
- First experiments with CUAs
- Build understanding of agent-based interaction with tender websites

Core Tasks

- Research available agents
- Select different agents per team
- Set up browser / GUI environments
- Run experiments (document downloading)
- Analyze behavior and limitations

Goals

- Initial agent implementations
- Hands-on experience with GUI agents
- Comparative insights:
 - Manual approach (doc. downloading)
 - CUA-based approach (doc. downloading)
(Requires a manually annotated test dataset)



3. Team Setup

Team Composition

- 4 students (Germany – TU Clausthal / Ostfalia)
- 3 students (Romania – UBB)
- 2 high school students

Collaboration Strategy

- Subteams work independently
- Teams compete to develop best solution
- Focus on exploration/experimentation
- Compare results and insights across teams

Team Structure

- Two subteams with 4 (5) members each
- Each team:
 - 2 TUC students
 - 1 (2) UBB student(s)
 - 1 high school student

Phase Workflow

- Phase 1: Different LLM approaches
- Phase 2: Different agent systems
- Phase 3: Independent final system

4. Project Roadmap

Phase 1

Automatic LLM-based Scraper Generation

- LLM-based document scraper code generation (at runtime)
- Evaluation dataset
- Feedback loops
- Safety & guardrails

Weeks 1-4

Phase 2

Computer Use Agent (CUA) Architectures

- GUI-based agents
- Interaction loops
- Parallel execution
- Cost analysis

Weeks 5-8

Phase 3

System Integration and Deployment (Integrated Service)

- Cascaded pipeline
- Fallback logic
- REST API
- Deployment

Weeks 9-12



4.1 Phase 1 – LLM-based Scraper Generation

Objective

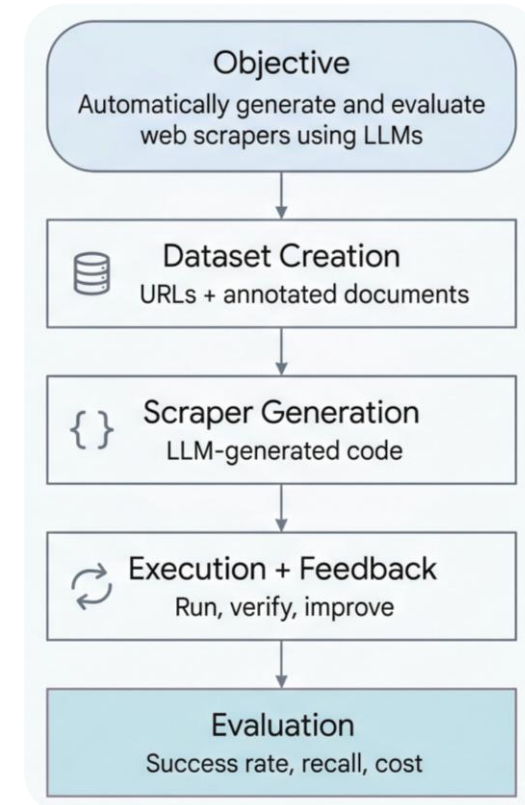
Automatically generate and evaluate scrapers using LLMs

Core Tasks

- Build a diverse evaluation dataset (URLs + expected documents) (both subteams should work on that together)
- Manual annotation of document counts and types
- Generate scraper code using different LLMs
- Execute generated code automatically
- Implement feedback loop (errors, states, success signals)

Exploration

- Compare multiple LLMs (*GPT, Claude, Gemini, etc.*)
- Evaluate prompting strategies and feedback mechanisms



4.1 Phase 1 – Evaluation and Safety

Evaluation Metrics

- Success rate of scraper execution
- Document recall (downloaded vs. expected)
- Runtime and latency
- API and compute cost

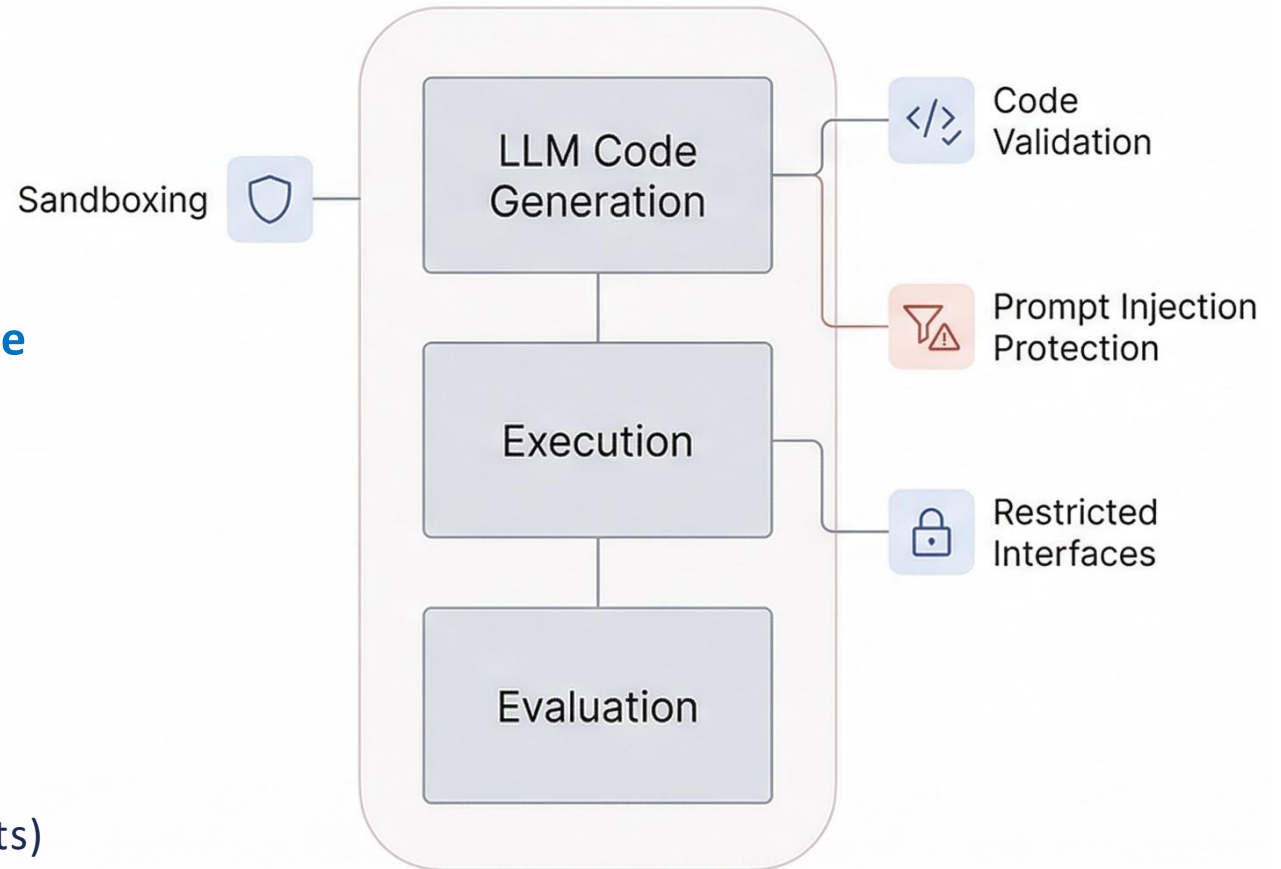
Safety and Guardrails for LLM-Generated Code

Key Risks

- Execution of unsafe generated code
- Prompt injection from web content
- Unauthorized system access

Safeguards

- Sandboxed execution (restricted environments)
- Code validation before execution etc.



4.1 Phase 1 – Deliverables

Software Artifacts

- Python library for automated scraping
- Input: URLs → Output: downloaded **documents**

Evaluation Results

- Benchmark across models and approaches
- Performance and cost analysis

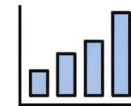
Additional Output

- Annotated evaluation dataset
- Security concept and guardrails



Software Artifacts

Python library for automated web scraping



Evaluation Results

Benchmark across models and approaches



Additional Outputs

Annotated evaluation dataset

4.2 Phase 2 – Computer-Use Agents (CUA)

Objective

Explore agent-based interaction with websites

Core Tasks

- Evaluate GUI-based agents on websites
- Loop: screenshots → actions → new states
- Explore multiple CUA frameworks (open + proprietary)

Technical Questions

- Browser automation vs. full GUI environments
- Parallel execution of multiple agents
- Scalability and robustness



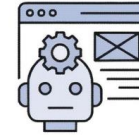
4.2 Phase 2 – Evaluation and Deliverables

Evaluation

- Success rate on evaluation dataset
- Number of interaction steps
- Runtime and cost per task
- Comparison to results from phase 1

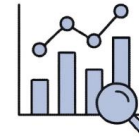
Deliverables

- CUA-based **document** scraping system
- Benchmark results of different CUA approaches
- Cost-performance analysis



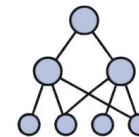
Agent System

GUI-based web interaction agents
URL → automated interaction and extraction



Evaluation Results

Success rate and task completion
Steps, runtime, and cost analysis



System Insights

Comparison of agent approaches
Scalability and robustness findings

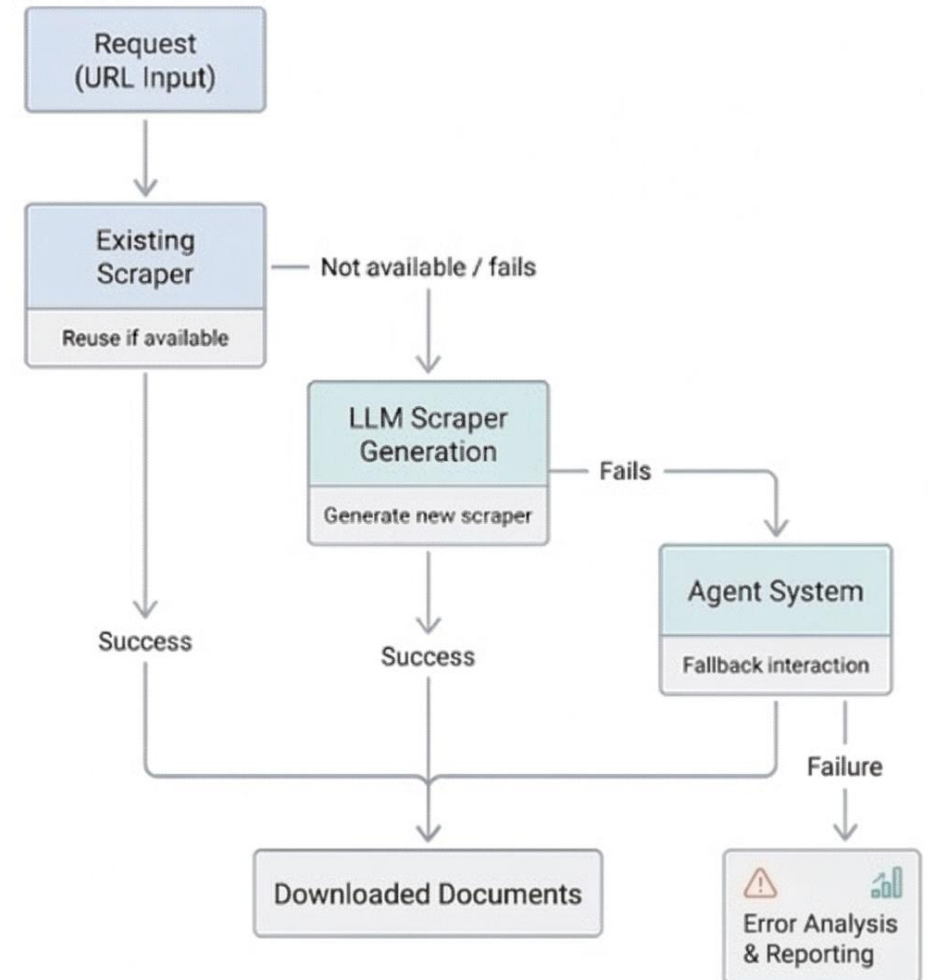
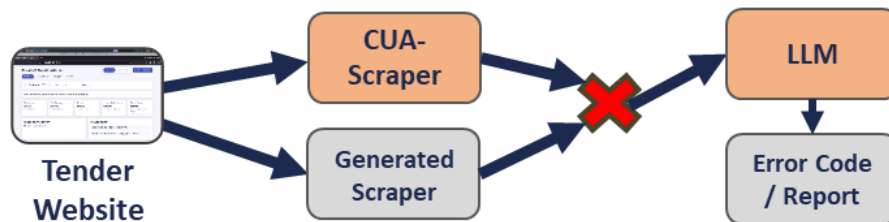
4.3 Phase 3 – Integrated System

Objective

Combine all approaches into a unified system

Core Tasks

- Reuse existing scrapers when available
- Generate new scrapers if needed
- Use CUA approach as fallback
- Automatic error analysis and reporting
- Support versioning of documents (periodic checks)



4.3 Phase 3 – System and Service Design

System Components

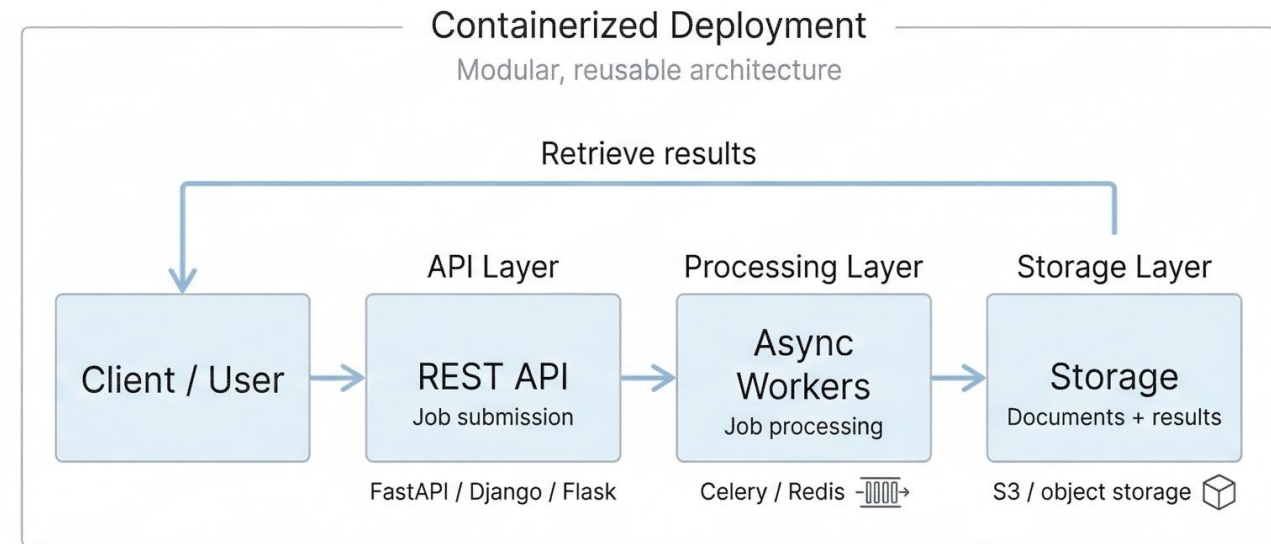
- REST API for job submission (e.g., *FastAPI, Django, Flask*)
- Asynchronous job processing (e.g., *Celery+Redis*)
- Persistent storage for results (e.g., *S3*)

Example Workflow

- POST URLs → receive job ID
- Query status via API
- Retrieve results (e.g., document links)

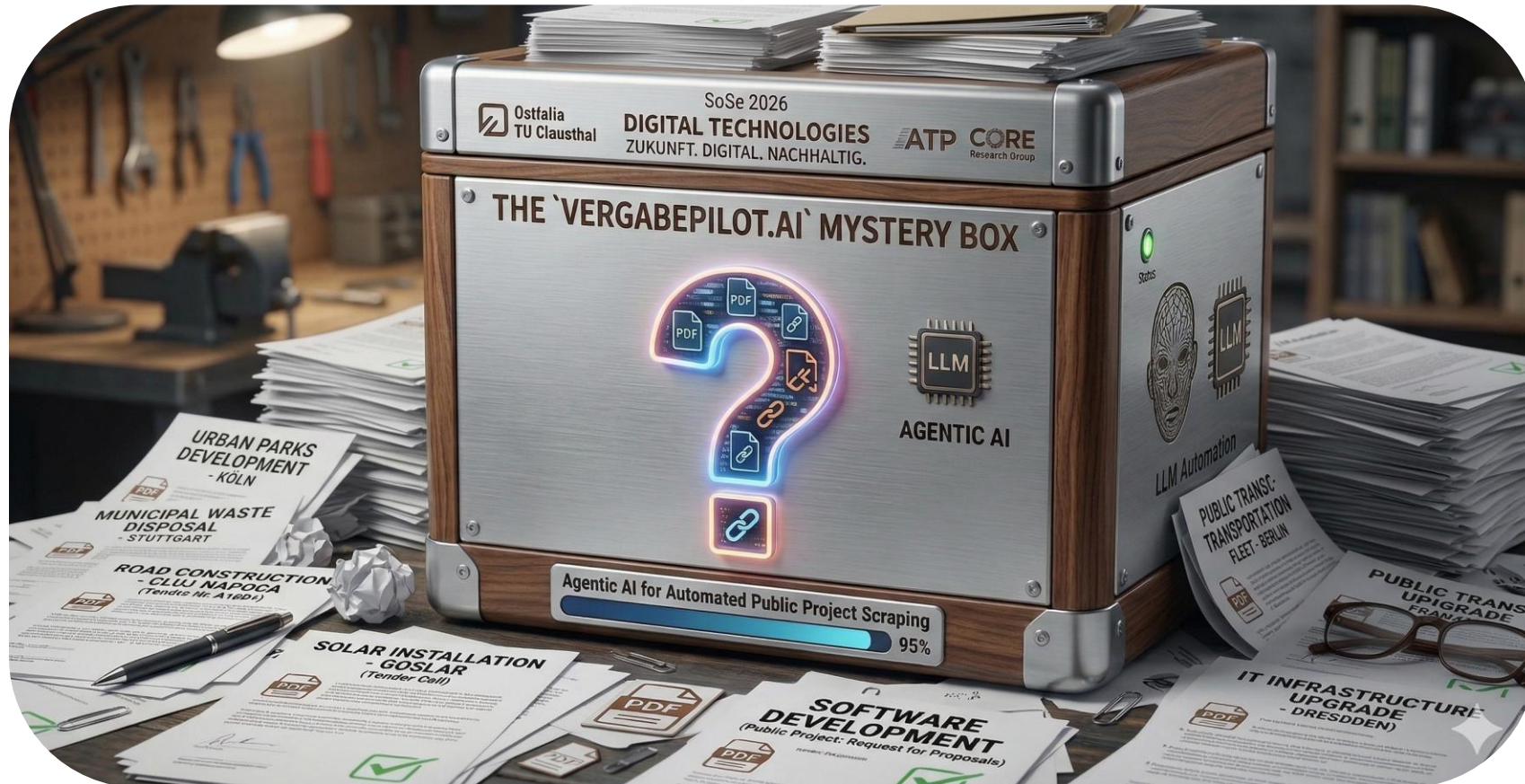
Deployment

- Containerized system (e.g., *Docker Compose*)
- Modular and reusable architecture



We will test your (blank) service with a hidden dataset at the end!
(for robustness, document download success rate, runtime, API cost)

4.4 Phase 4 – Secret Challenge



Will be revealed once you fully complete the first three phases!

5. Technical Setup

Model & API Access

- Ciconia *OpenRouter* API keys (multi-model)
- Flexible choice of LLMs
- Fallback: direct proprietary APIs
- Starting Budget: 50\$

Datasets

- Large dataset (thousands of URLs of project websites)
- Annotated evaluation datasets (student-built)

Development

- GitHub for version control
- Collaboration via shared repository
- Use private repositories

Infrastructure & Tools

- University compute resources (GPUs)
- Support for local open-source models
- Scrum board (*Trello*)

6. Grading

50%

Created code artifacts, conducted and documented evaluations

50%

Presentations

7. SCRUM Setup

- **Sprints** last **2 weeks**
- Each sprint begins with **Sprint Planning**
- Each sprint ends with a **Sprint Review** and **Retrospective**
- **Roles:** ***Product Owner*** and ***Scrum Master*** → suggestion: rotate every **2 weeks**
- The **Product Owner** is responsible for user stories and requirements management
- Every task is tracked; at the end, an artifact is created for each task
- The **daily** (stand-up) is conducted by your team independently
- Attendance is **mandatory for all meetings**
- First absence = **warning**; second absence = **removal from the project**
- **Alternating meetings (for each subteam):**
 - *Sprint planning and review (every 2 weeks, 30 min) / Weekly (in between, 15 min)*

8. Retrospective

What did you learn from this week (Hackathon + Cluj)?

- What insights did you gain about the project or tech?
- What experiments worked / did not work?
- What would you do differently if we started again?

Where did you struggle?

- When were things unclear (tasks, goals, communication)?
- What felt unstructured or inefficient?
- What did you struggle with the most?

9. Sprint Planning

Each team should create a board according to the following schema (e.g., **Trello**)

User Stories	Backlog	Sprint Backlog	In Progress	In Review	Done
All user stories provided by product owner	All tasks derived from user stories left in the backlog Ordered by priority (high to low)	Overview of current sprint			
		All tasks not touched by the team yet	All task worked on right now One per team member	All tasks done but not reviewed yet	All reviewed and approved tasks